

User Stories for Agile Software Development

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January 17, 2020

Agile software development

Manifesto (2001)

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

User stories

For agile software development — emphasis on communication and feedback

- features
- planning
- estimation
- specification
- testing
- implementation
- progress
- documentation

User stories – Why Who What

- a title
- a narrative
 - in order to (purpose – why),
 - as a (user role – who),
 - I can (action on object – what)

- brief notes during brainstorming
- acceptance criteria
(later, during specification)

INVEST

- Independent
- Negotiable – conversation
- Value to user and buyer
 - visible (understandable) to user
 - no technical jargon
- Estimable
 - domain knowledge – conversation
 - tech knowledge – spike
 - too big – split
- Small
- Testable

User roles

Team workshop for gathering user roles

- brainstorming
- generate initial set
- consolidate, refine

Collecting user stories

Team workshop for low-fidelity prototype

- start with the home page
 - each piece of information on the page
- next page after a click
 - each piece of information on the page
 - may be different for different user roles
- ...

- quantity more important now
 - may allow epics
- stop when you have got more than enough

Backlog

- a list of user stories
- priority
 - infrastructure need
 - technical dependencies
 - broadness of users
 - a few important users
 - collectively with customer/stakeholder
- status
 - scheduled, specified, working-on, done

Estimation

- story points
 - used only within the project
 - 1, 2, 3, 5, 8, 13, ∞ , ?
 - 13 — a single developer can finish within an iteration
 - constant during project
 - * the velocity may change
- planning poker

Planning the next iteration

- how many story points?
- which stories?
 - split stories that are too big
 - spikes
- who works on what?
- tasks
 - can be technical
 - can be independent of stories
 - ideal time

Specification of user stories

- collective
 - whole team
 - three amigos
 - one write, others review
- use examples
 - happy cases
 - edge cases
 - exceptional cases
- translate into acceptance tests
 - as living documentation
 - context – action – effect

During iteration

- do not change the plan
- daily stand-up meetings
- development, tests
 - outside-in, inside-out
- a story is done only when all tests are green
- Git
 - branching policy
- pair programming, code review

End of iteration

- milestones, releases, ...
- demo
- feedback
- celebrate
- velocity, planned and actual
- retrospective
- take the chance to change/adjust
- plan next iteration

Start now

- user roles
- low-fidelity prototype
- backlog
- estimate
- plan the first iteration
- try to specify user stories